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Overview

This project develops and validates a Process Design Kit (PDK) for a four-mask NMOS transistor process intended for a quantum-dot device platform. Before committing to silicon, the complete fabrication flow and the resulting device were simulated end-to-end in Synopsys Sentaurus TCAD: the process steps were modeled to build the physical structure, a design-of-experiments swept the key process parameters, and device simulations extracted the electrical characteristics to confirm the transistor meets its targets.

- **Process simulation (SProcess).** Modeled the four-mask flow — oxidation, implantation, diffusion, etch, and metallization — to construct the device cross-section and its doping profiles (Figures 1–2), including the source/drain junctions, channel, and thin gate oxide.
- **Design-of-experiments (Sentaurus Workbench).** Parameterized the flow and swept key process variables (gate-oxide thickness, channel and source/drain doping, field-oxide and gate parameters) to study their effect on device behavior (Figure 3).
- **Device simulation & extraction (SDevice).** Ran electrical simulations on each generated structure and extracted the transistor figures of merit — threshold voltage (V_t), subthreshold swing (SS), transconductance (g_m), and drive current (I_d) — across the full parameter sweep (Figure 4).

The flow successfully produced a working NMOS transistor across the swept process corners, with a well-formed channel and source/drain junctions and a controlled thin gate oxide. SDevice extracted consistent device parameters — threshold voltage, subthreshold swing, transconductance, and drive current — confirming that the PDK's process targets yield a functional device. The simulated PDK was deemed a success and now anchors the mask design and fabrication phase.

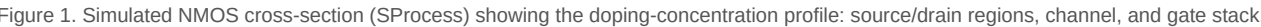


Figure 2. SDevice extraction results across the parameter sweep — threshold voltage, subthreshold swing, transconductance, and drive current for each process split.

Synopsys Sentaurus TCAD (SProcess process simulation, SDevice device simulation, Sentaurus Workbench DoE); semiconductor process integration; MOSFET device physics; parameter extraction (V_t , SS , g_m); Process Design Kit (PDK) development; design-of-experiments and process centering.